- Problem Statement ID – GEO-04
- Problem Statement Title – Bi-Temporal Satellite Change Detection for Rapid Urban Encroachment & Disaster Impact
- Team Name – HACK4CHANGE
- Team Leader Name – DIVISHA PANJWANI
- Members Names – TANISHA CHOPADE, BHAKTI DESHMUKH, SHRAWANI AKKANAVARU

TERRAWATCH AI

From satellite changes to response priorities: finds what changed in any place in India, explains which changes are false alarms, ranks where people and roads are affected, and hands field teams a report in their language.

Any place: fetch or upload → **Dual-engine AI detection** → **Explained false alarms** → **People, facilities & roads** → **Priority queue & field report**

BUILT | BUILT | BUILT | BUILT | BUILT

What it does

- Fetches free Sentinel-2 images for any place and date, or takes uploaded GeoTIFFs
- Outputs GIS polygons with area (m^2), magnitude, centroid, severity and estimated type
- Ranks changes by severity, people, nearby hospitals/schools and blocked roads

How it solves the problem

- Replaces slow manual comparison of wide areas with automated triage
- Rejects clouds, seasonal crops and shadows — and shows why
- Flags villages that may be cut off from a hospital; one-click report for field teams

What makes it unique

- Resolution-aware dual engine: Siamese CNN (0.5 m) + Sentinel-2 U-Net (10 m)
- Explainable at every step — nothing silently hidden, no AI-invented text
- Disaster + Prevention modes; reports in English, हिंदी and मराठी

TECHNICAL APPROACH

Technologies

AI / ML
PyTorch · TinyCD Siamese CNN (LEVIR-CD) · U-Net trained on OSCD

Geospatial
Rasterio · GeoPandas · Shapely · osmnx · NetworkX

Data
Sentinel-2 L2A + SCL (Planetary Computer) · OpenStreetMap · WorldPop

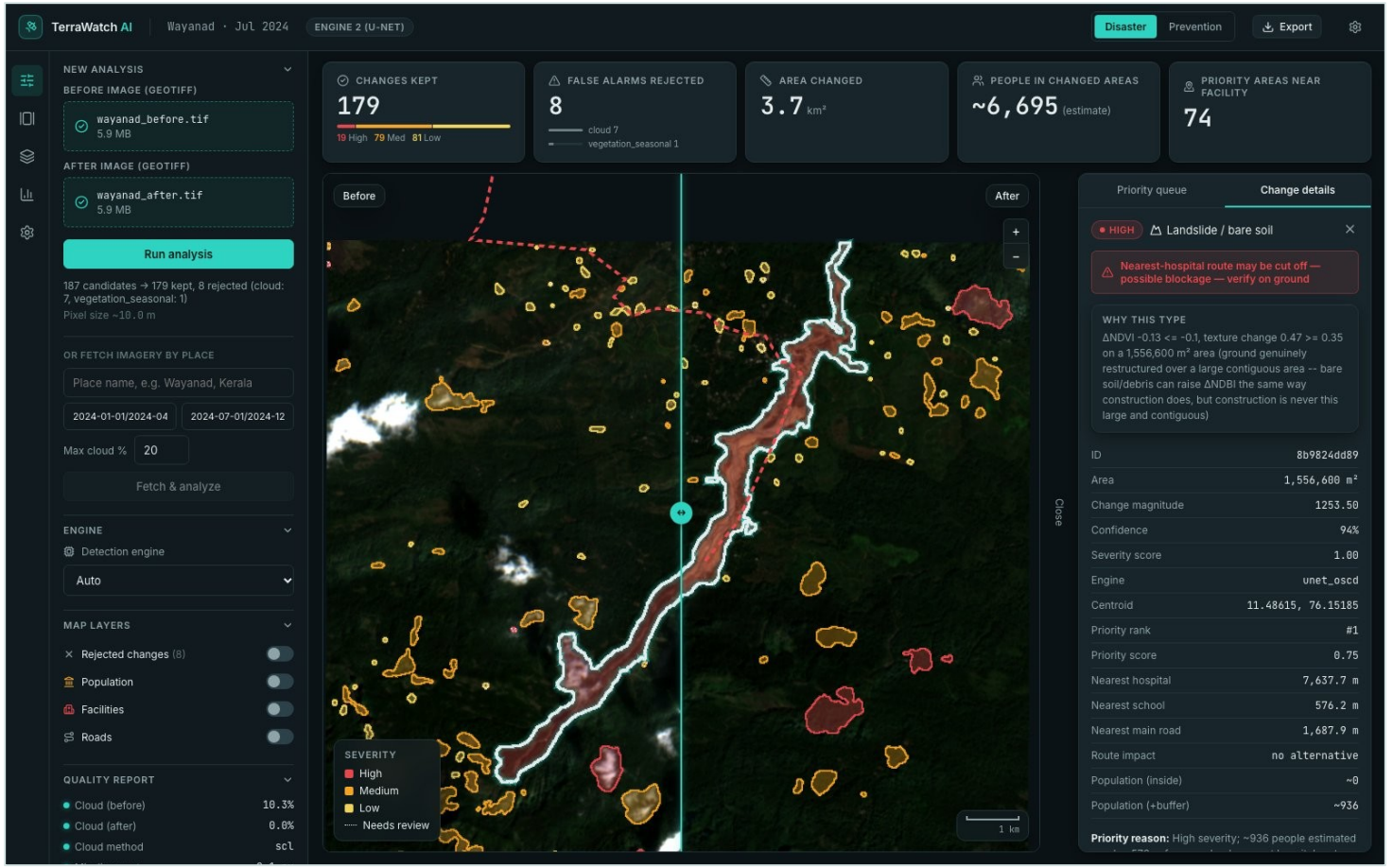
Backend
Python 3.11 · FastAPI · SQLite · WeasyPrint · 99 tests + CI

Frontend
React · TypeScript · MapLibre GL

Methodology: the pipeline

1. **Get imagery**
   Place name or drawn box → lowest-cloud Sentinel-2 pair, radiometrically harmonised; or upload GeoTIFFs
2. **Align & quality check**
   One pixel grid; cloud %, pixel shift, resolution, scene-dominance warning
3. **Detect**
   ≤2 m RGB → TinyCD Siamese CNN · 10 m Sentinel-2 → U-Net · tiled change-probability map
4. **Clean & vectorize**
   Threshold + morphology → Rasterio polygons: area m², magnitude, centroid, severity
5. **Explain false alarms**
   Cloud (SCL), seasonal vegetation, shadow, too small, misalignment → rejected with reason
6. **Impact & routes**
   People (WorldPop), nearest hospital/school/road, road-graph check for cut-off villages
7. **Deliver**
   Priority queue · map · GeoJSON/GeoPackage · PDF/mobile report in EN/हिंदी/मराठी

TECHNICAL APPROACH: PROTOTYPE

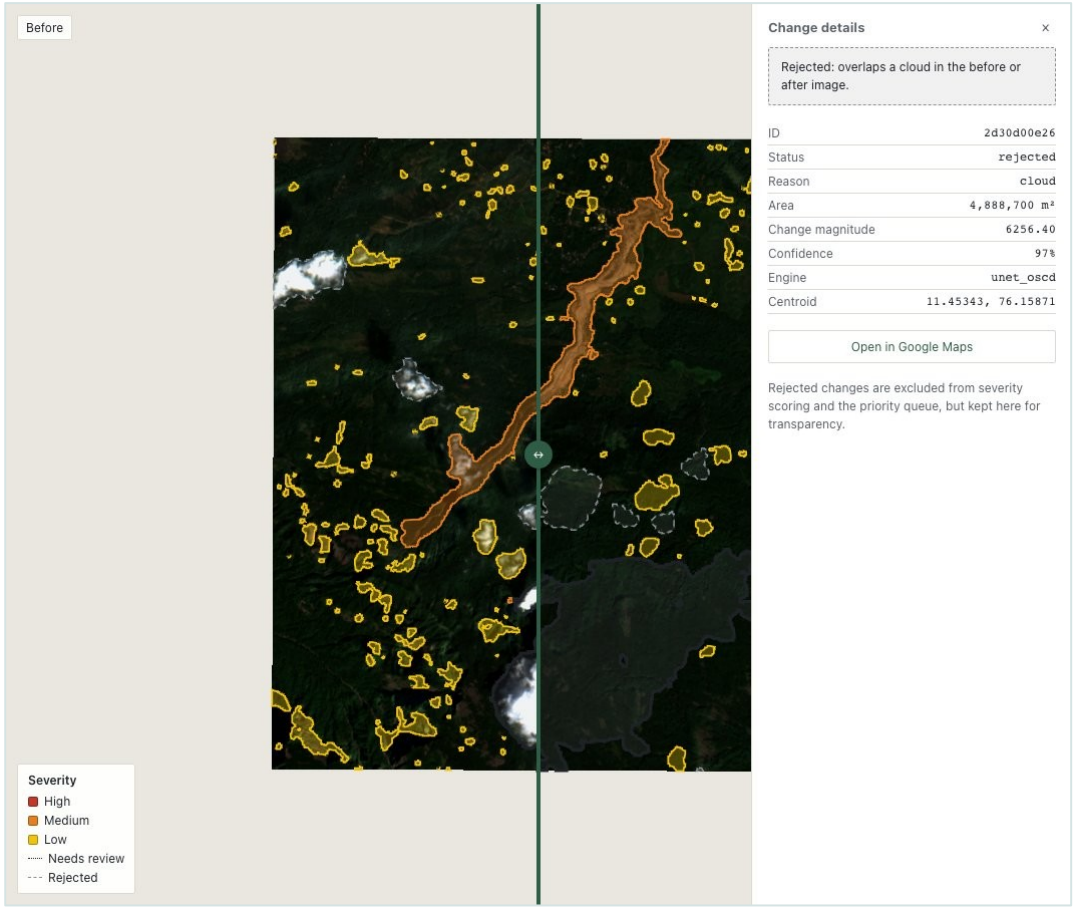


Wayanad landslide (Jul 2024): ranked #1, High severity, 1.56 km² — nearest-hospital route may be cut off

Running today

- Fetch any place in India by name or drawn box
- 4 real Indian case studies, one click
- Auto engine selection + quality report
- False alarms rejected with a visible reason
- People, hospitals, schools, roads per change
- Blocked-route check for cut-off villages
- Priority queue · Disaster & Prevention modes
- PDF/mobile field report in EN/हिंदी/मराठी

TECHNICAL APPROACH: FALSE ALARMS



A cloud in the before image, rejected with its reason (97% confidence) — not silently hidden

Every change is KEPT or REJECTED with a reason

Reason	Evidence used
cloud	Sentinel-2 SCL cloud mask in either image
vegetation_seasonal	NDVI change with no built-up / water / soil change
shadow_illumination	Brightness-only change, little structural change
too_small	Below minimum area
misalignment_edge	Thin edge slivers from an image shift

0.38 → 0.73

Precision

labelled test scene, all real changes still found (F1 0.55 → 0.83)

7

Clouds rejected

Wayanad: 187 candidates → 179 kept, 8 rejected

3.0 px

Shift detected

misaligned test image caught by the quality check

TECHNICAL APPROACH: MEASURED RESULTS

Ablation on held-out test sets (from our evaluate.py)

Engine / dataset	Stage	Precision	Recall	F1	False-positive rate
TinyCD · LEVIR-CD (128 pairs, 0.5 m)	Raw model	92.9%	89.9%	**0.914**	0.4%
	+ false-alarm filter	93.1%	87.7%	**0.903**	0.4%
U-Net (Engine 2) · OSCD (10 cities, 10 m)	Raw model	19.6%	93.5%	**0.324**	20.9%
	+ false-alarm filter	20.0%	92.6%	**0.329**	20.2%
Spectral baseline · OSCD	Raw	15.8%	59.2%	**0.250**	17.1%
	+ false-alarm filter	21.9%	53.4%	**0.311**	10.4%

What this shows, honestly: the high-res engine is strong (F1 0.914). On 10 m Sentinel-2, our U-Net catches over 92% of real change; the filter cuts the baseline's false-positive rate from 17.1% to 10.4% and makes every rejection explainable. Raising U-Net precision with longer training and Indian disaster data is our next modelling step.

OVERNIGHT DEVELOPMENT (AFTER ROUND 2)

Any place in India
Fetch Sentinel-2 by place name or drawn box; 4 real case studies (Wayanad, Sikkim GLOF, Himachal floods, Pune)

Blocked-route checker
Road-graph check: 10 changes in Wayanad may cut villages off from a hospital — incl. the #1 landslide

Field report in 3 languages
One-click PDF + mobile page in English, हिंदी, मराठी — computed values only, no AI-invented text

Satellite-data bug fixed
ESA's 2022 processing change made old vs new images look 'all changed': false 30.4 km² polygon → 0.3 km² (100× smaller)

Severity & ranking fixed
0 / 2 / 177 → 19 / 79 / 81 (High/Med/Low); landslide back to High and #1

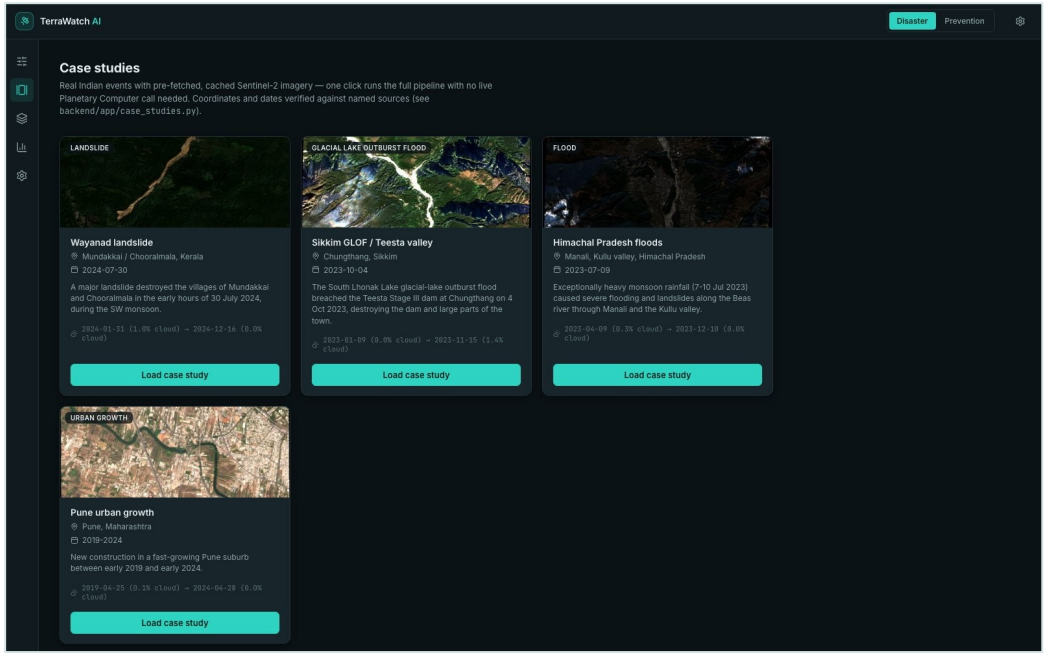
Product-grade UI
Dark GIS interface, KPI strip, one-click demo, draw-a-box with size safeguard

Quality
Tests 69 → 99 passing · CI on every push · 4 UI bugs and 2 pipeline bugs found and fixed

Documentation
README requirement map, architecture, evaluation, model card, user guide, Wayanad case study

All work timestamped in git after 8:00 PM, 9 Oct · full log with every number: docs/OVERNIGHT_DEVELOPMENT.md

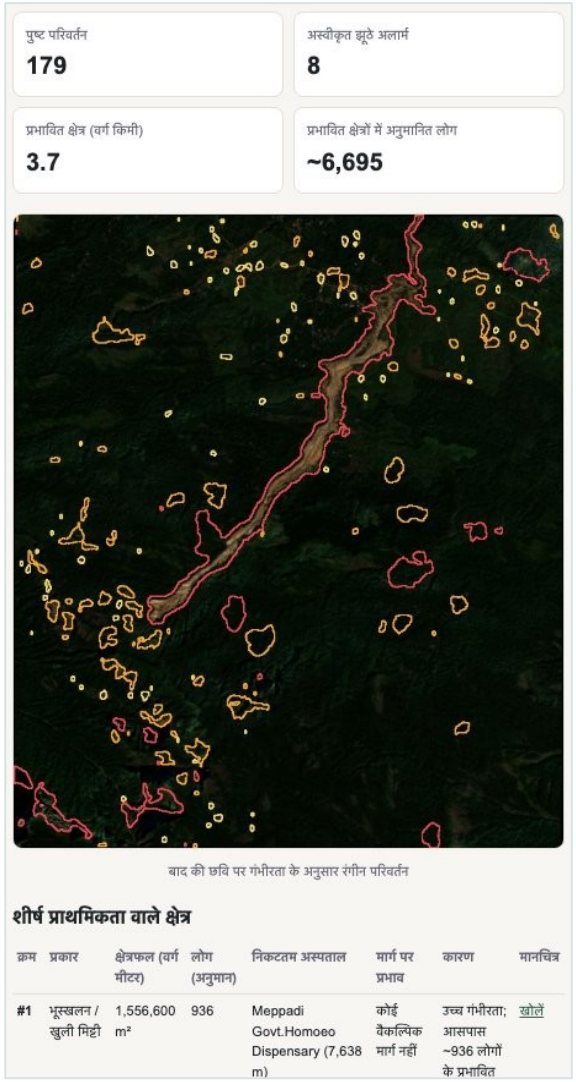
OVERNIGHT DEVELOPMENT: IN THE APP



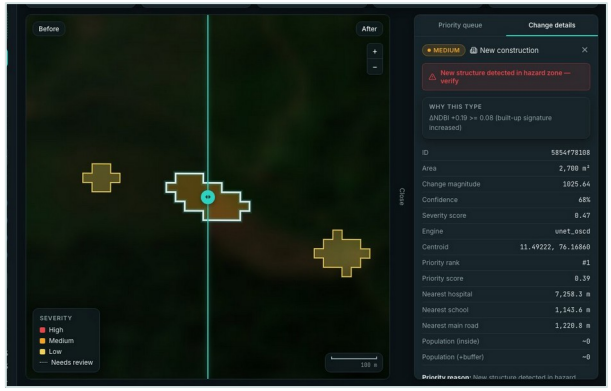
Case studies: real Indian events, one click



Pune 2019 → 2024 after the fix
Images from before and after ESA's 2022 change are now harmonised (−1000 offset), plus a safeguard that warns if one change dominates the scene.



Hindi field report (mobile)



Prevention: new structure in a hazard zone

10

Possible cut-offs

changes where the nearest-hospital route has no alternative (Wayanad)

99

Tests passing

up from 69 at Round 2, CI on every push

FEASIBILITY AND VIABILITY

Why it's feasible

Working end to end
Any place in India, 4 case studies, 99 automated tests

Free, open data
Sentinel-2 (~5-day revisit), OpenStreetMap, WorldPop

Runs on a laptop
Pretrained Siamese CNN + lightweight U-Net

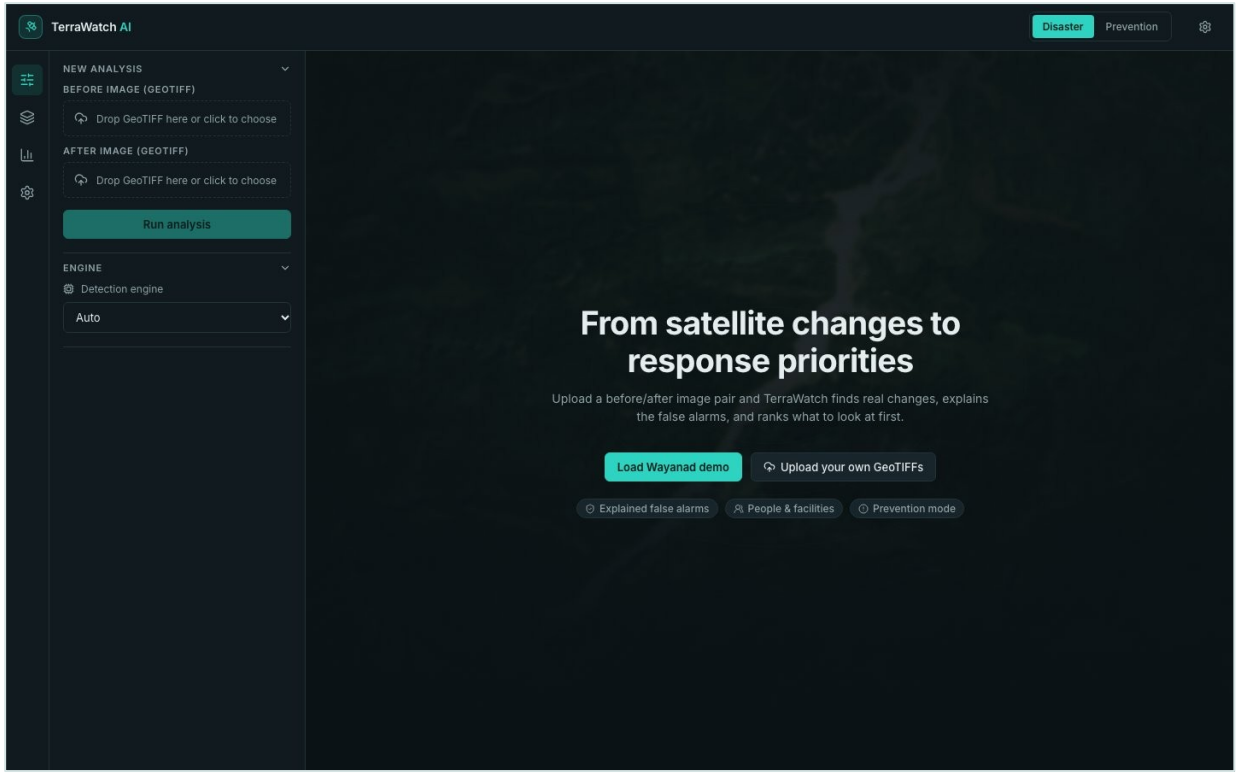
Fits existing workflows
GeoJSON / GeoPackage for QGIS & ArcGIS; PDF/mobile reports

Challenges and how we handled them

Challenge	What we built
False alarms: clouds, seasons, shadows	Explained filter with 5 rejection rules; SCL cloud mask
One model fails across resolutions	Auto engine selection; trained our own Sentinel-2 U-Net
Old vs new satellite images differ	Radiometric harmonisation for ESA's 2022 change + dominance warning
Misaligned or cloudy input	Quality report: cloud %, pixel shift, resolution
Change ≠ damage	Severity = change strength; estimates labelled; 'verify on ground'

Viability: for SDMA / district disaster cells, municipal planners and NGOs · near-zero data cost · laptop or small cloud server

IMPACT AND BENEFITS



TerraWatch AI — from satellite changes to response priorities

~6,695
People (estimate)
in changed areas

74
Priority areas
near a facility

10
Possible cut-offs
no alternative route

Emergency responders
Ranked list of where to go first, with people, nearest hospital and possible cut-off roads

Urban planners
Prevention mode flags new construction inside flood- and landslide-prone zones

Communities & field teams
Reports in English, Hindi and Marathi with Google Maps links; every flag marked for verification

"Today's encroachment is tomorrow's disaster."

IMPACT AND BENEFITS: WHAT'S NEXT

Next weeks	Scale months	Deploy pilot
All-weather radar Sentinel-1 SAR sees through monsoon clouds and at night	**Watchlists & alerts** Auto-analyse every new satellite pass; alert district teams	**Pilot with SDMA / NGOs** Start with Kerala & Maharashtra hazard districts
Terrain-aware rules Elevation & slope: landslides need slopes, floods need low ground	**Early-warning signs** Time series: slope forest loss, construction in floodplains	**Field app** Offline-capable mobile report and verification
Better Sentinel-2 model Longer training with Indian disaster data for higher precision	**More languages** Tamil, Malayalam, Bengali and more for field reports	**Municipal monitoring** Track construction near Pune's rivers and hills
Learns from the field One-tap verify / false-alarm feedback retrains the models	**Plug into existing tools** Feeds for QGIS, ArcGIS and government GIS portals	**Open & low-cost** Open-source core on free data; small cloud server

RESEARCH AND REFERENCES

Datasets & models

1. Chen, H. & Shi, Z. (2020). A spatial-temporal attention-based method and a new dataset for remote sensing image change detection (LEVIR-CD). Remote Sensing.
2. Daudt, R. C., Le Saux, B., Boulch, A. & Gousseau, Y. (2018). Urban change detection for multispectral Earth observation using CNNs (OSCD). IGARSS.
3. Codegoni, A., Lombardi, G. & Ferrari, A. (2023). TINYCD: a (not so) deep learning model for change detection. Neural Computing and Applications.
4. Daudt, R. C., Le Saux, B. & Boulch, A. (2018). Fully convolutional Siamese networks for change detection. IEEE ICIP.
5. Ronneberger, O., Fischer, P. & Brox, T. (2015). U-Net: convolutional networks for biomedical image segmentation. MICCAI.
6. Wayanad event facts: ISRO/NRSC via The News Minute; International Consortium on Landslides (Mundakkai report).

Data sources, tools & repo

- Copernicus Sentinel-2 L2A incl. SCL (ESA), via Microsoft Planetary Computer
- OpenStreetMap contributors (Overpass, osmnx)
- WorldPop population grids
- PyTorch, Rasterio, GeoPandas, FastAPI, MapLibre GL JS, WeasyPrint
- GitHub: github.com/divishapanjwani/terrawatch
- Full overnight log: docs/OVERNIGHT_DEVELOPMENT.md